

Reproducing a DDPG-Based Stock Trading Strategy

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1) Problem and Goal

Problem: Daily stock trading is a noisy sequential decision-making task.

Project objective:

- Reproduce Liu et al. (2018) DDPG stock-trading setup.
- Use the same Dow Jones split and compare against:
 - DJIA benchmark
 - Min-variance portfolio baseline
- Test whether TD3 improves stability and performance over DDPG.

2) Environment and MDP

State:

$$s_t = [b_t, p_t^{(1..28)}, h_t^{(1..28)}]$$

where b_t is cash, p_t are prices, and h_t are holdings.

Action: integer trades per stock

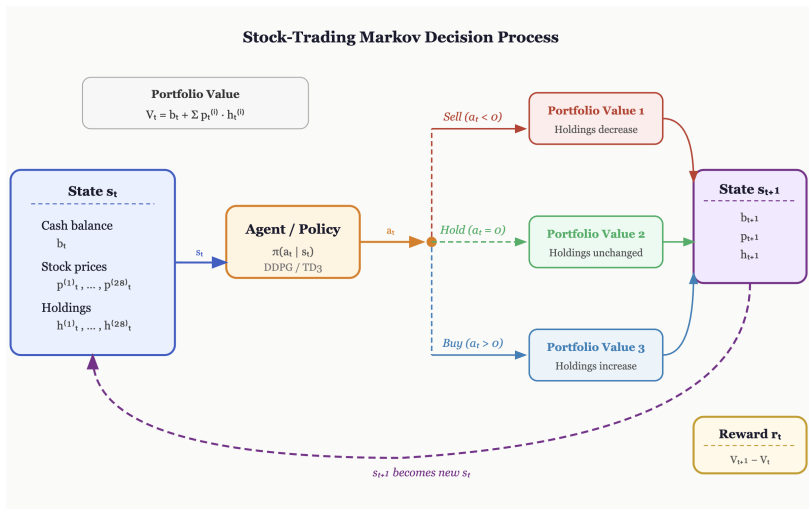
$$a_t = [a_t^{(1)}, \dots, a_t^{(28)}], \quad a_t^{(i)} \in \{-5, -4, \dots, 0, \dots, 4, 5\}$$

with long-only and cash constraints.

Reward:

$$r_t = V_{t+1} - V_t, \quad V_t = b_t + \sum_{i=1}^{28} p_t^{(i)} h_t^{(i)}$$

MDP Agent-Environment Interaction



3) Methods and Experimental Setup

Algorithms

- DDPG (paper baseline): deterministic actor-critic, replay buffer, target networks.
- TD3: twin critics + delayed actor updates + target smoothing.

Training and evaluation protocol

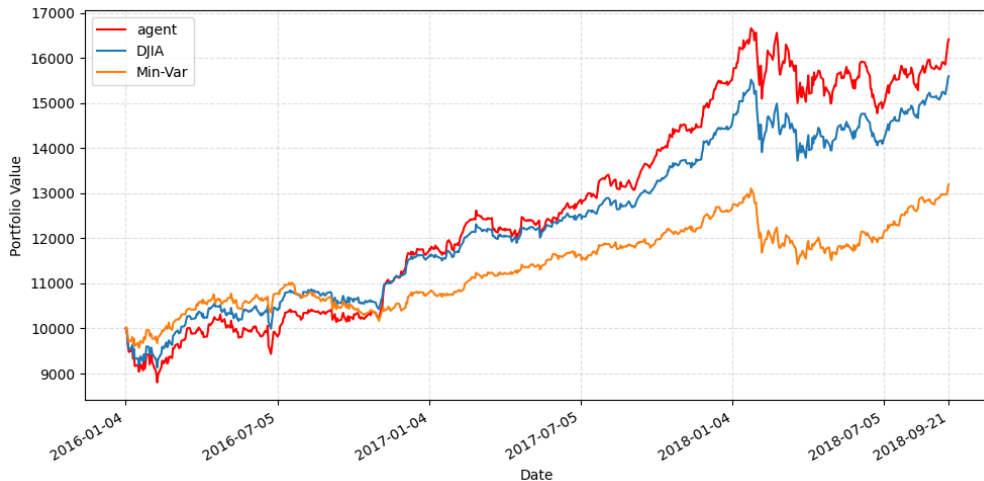
- Stable-Baselines3 implementation.
- Gymnasium for defining the custom stock trading environment
- Data split: train (2009–2014), validation (2015), test (2016–2018).
- Hyperparameter tuning with Optuna: 10 trials per algorithm.
- Final evaluation: 30 episodes, 3 seeds (42, 43, 44).

4) Main Results (Test Period 2016–2018)

Metric	DDPG	TD3	Min-Var	DJIA
Final Portfolio Value	\$16,141	\$16,919	\$13,194	\$15,595
Annualized Return	19.25%	21.31%	10.74%	17.76%
Annualized Std.	12.78%	12.67%	9.37%	11.74%
Sharpe Ratio	1.51	1.68	1.15	1.51

- DJIA is a core benchmark in the original work and in our reproduction.
- DDPG slightly beats DJIA on return, with similar Sharpe.
- TD3 achieves the strongest overall risk-adjusted performance.
- TD3 performs better on average, but with standard deviation considered, it remains close to DDPG.

Sample Trade Run (DDPG, Seed 42)



5) COVID-19 Stress Test (Robustness Check)

Setup: Train on 2010–2019, trade on volatile 2020 (COVID), same Dow Jones universe, best reproduced hyperparameters, 500 episodes, 3 seeds.

Metric	DDPG	TD3	Min-Var	DJIA
Final Portfolio Value	\$10,590	\$9,786	\$10,477	\$10,534
Annualized Return	5.93%	-2.15%	4.79%	5.36%
Sharpe Ratio	0.18	-0.04	0.20	0.15

- RL does not consistently outperform DJIA and min-variance in this volatile regime.
- This questions the original claim that DDPG outperforms classical baselines, since performance degrades under volatile market conditions.

6) Interpretation, Limits, and Future Work

What we reproduced

- RL agents outperform min-variance baseline in this setting.

What differed from original paper

- Reproduced DDPG score is below the reported original DDPG result.
- Likely causes: smaller compute budget and less extensive tuning.

Next steps

- Add transaction costs and slippage.
- Increase training/tuning budget and test additional market periods.
- Compare against newer methods (e.g., SAC).

7) Takeaways

- The original DDPG stock-trading trend is **partially reproduced**.
- **TD3 outperforms DDPG on the original split**, but not robustly across all regimes.
- Reinforcement learning can be promising for portfolio control, but realistic assumptions and robust evaluation are critical.

Thank you.